

Lab Exercise: Connecting the Course to Your Research

Introduction to Applied Regression Analysis — ICPSR 2026

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Overview

For this exercise, think about a research question you are working on or would like to explore. You do not need a polished project — even a rough idea is fine. Take 15–20 minutes to work through the questions below individually, then we will discuss as a group.

Part A: From Question to Model (5 minutes)

Write down, in plain language:

1. What is your research question?
2. What outcome are you trying to explain (your Y)?
3. What is the main factor you think matters (your X)?
4. What kind of data would you use, and what does a single observation look like (a person, a country-year, a firm, etc.)?

Part B: Complications and Choices (5–10 minutes)

Pick **two** of the following prompts that feel most relevant to your project and write a few sentences on each.

- a. **Controls and confounders.** What other variables might you need to control for, and why? Are any of them potential confounders — variables that affect both X and Y ?
- b. **Functional form.** Do you think the relationship between X and Y is linear, or might you need a different specification (e.g., logged variables, polynomial terms, interactions, logit/probit)?
- c. **Panel structure.** If your data follow the same units over time, what challenges does that create? How might you address them?
- d. **Threats to inference.** What would make you worried that your estimate of the $X \rightarrow Y$ relationship is biased? What could you do about it?

Part C: What Would Make This Convincing? (5 minutes)

If you could do one thing to strengthen the analysis you have just sketched — collect better data, use a different model, find an instrument, run a specific diagnostic — what would it be, and why?

Group Discussion

After completing the exercise, we will discuss as a group:

- What was hardest about translating your research interest into regression terms?
- Where do you see the biggest gap between what we covered this week and what you would actually need for your project?
- Did anything we covered this week change how you think about a project you are already working on?